Formal Methods in Software Engineering

Credit Hours: 3

By: Muhammad Hamza
System Engineer/Lecturer

Introduction

- Muhammad Hamza
- Currently working as a Senior Software Engineer.
- Having more than 4 years of Industry Experience
- Area of Interest
  - Software Engineering
  - Software Testing & QA
  - Computer Networks & Security
  - AI/ML/DL/CV
  - LLMs/NLP/DM

Instructor's Introduction

- **MS Computer Science (LUMS)**
- **BS Computer Science (GCU)**
- **Strong Industry IT professional worked with International & Local Clients**
- **Have worked with Services based and product based companies in the IT sector.**



Course LOs

- **CLO1: Describe the costs and benefits of formal methods.**
- **CLO2: Construct formal models of sequential software systems.**
- **CLO3: Implement sequential software systems based on formal models.**
- **CLO4: Verify attributes of formal models.**
- **CLO5: Demonstrate formal correctness of simple procedure.**



Instructor's Teaching Method

- **Lecturing,**
- **Written Assignments**
- **Project**
- **Report Writing**



Instructor's Experience

- **Navigation & Communication Solutions Pvt Ltd (System Engineer)**
  - **Jan 2025 - Now**
- **System Heuristics (Software Engineer)**
  - Oct 2023 - Dec 2024
- **Rolus Tech (Software Engineer)**
  - June 2022 - Aug 2023



Grading Policy

Area of Course	*Weight*
(Home Assignments, Quizzes, Project, Presentations)	25%
Mid Term	35%
Final Exam	40%

Text & Reference Book(s)

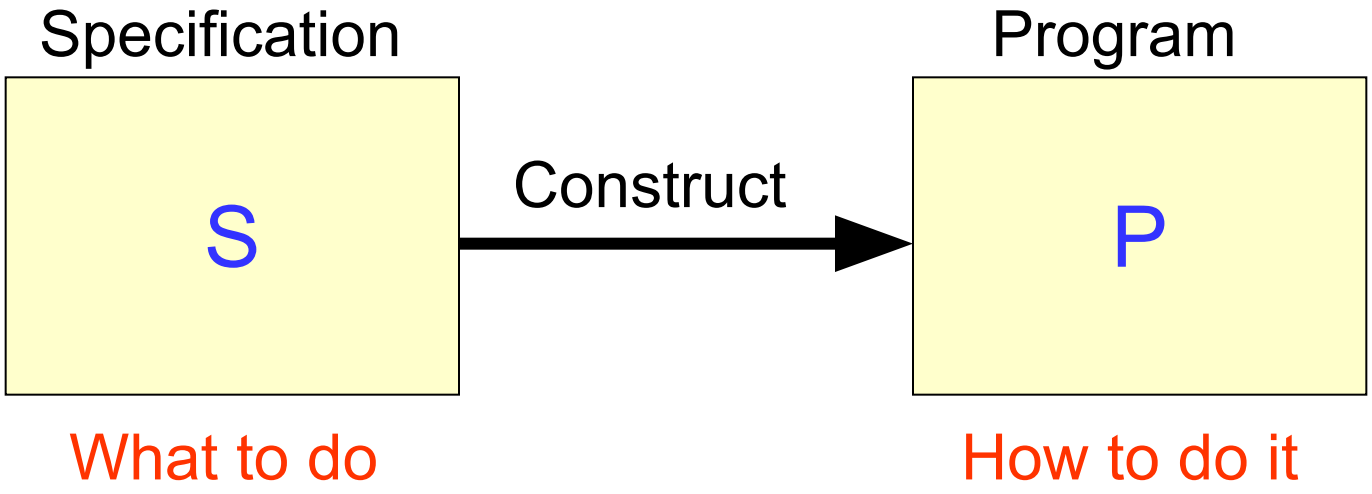
Primary Text Book(s)

- O'Regan, Gerard - Concise Guide to Formal Methods _ Theory, Fundamentals and Industry Applications-Springer International Publishing (2017)
- Modern Formal Methods and Applications Edited by HOSSAM A. GABBAR Okayama University, Okayama, Japan

Additional Reference Book(s)/Material

- Guide to the Software Engineering Body of Knowledge v4.0 EDITOR Hironori Washizaki Waseda University, IEEE Computer Society 2024 President-Elect, 2025 President

Problems in software development



- How to ensure that S is not ambiguous so that it can be correctly understood by all the people involved?
- How can S be effectively used for inspecting and testing P?
- How can software tools effectively support the analysis of S, transformation from S to P, and verification of P against S?

An example of informal specification:

"A software system for an Automated Teller Machine (ATM) needs to provide services on various accounts. The services include operations on current account, operations on savings account, transferring money between accounts, managing foreign currency account, and change password. The operations on a current or savings account include deposit, withdraw, show balance, and print out transaction records."

A better way to write the same specification:

"A software system for an automated teller machine (ATM) needs to provide services on various accounts.

The services include

1. operations on current account
2. operations on savings account
3. transferring money between accounts
4. managing foreign currency account,
5. change password.

The operations on a current or savings account include

1. deposit
2. withdraw
3. show balance
4. print out transaction records."

The major problems with informal specifications:

- Informal specifications are likely to be ambiguous, which is likely to cause misinterpretations.
- Informal specifications are difficult to be used for inspection and testing of programs because of the big gap between the functional descriptions in the specifications and the program structures.
- Informal specifications are difficult to be analyzed for their consistency and validity.
- Information specifications are difficult to be supported by software tools in their analysis, transformation, and management (e.g., search, change, reuse).

A possible solution to these problems:

Formal Methods!!!

Formal methods for the problems

What is formal methods?

Formal methods = Formal Specification

+

Refinement

+

Formal Verification



Set theory, logics, algebra, etc.

What are Formal Methods?

- Definition

  Mathematically based techniques for the specification, development and verification of software and hardware systems.

  Can determine if:

  - Specifications are satisfied.
  - Implementation of a system is correct.
  - Proving properties of a system.

What are Formal Methods?

- Formal Language Specification Formal Methods centred around a notation known as a formal specification language.

Formal Semantics

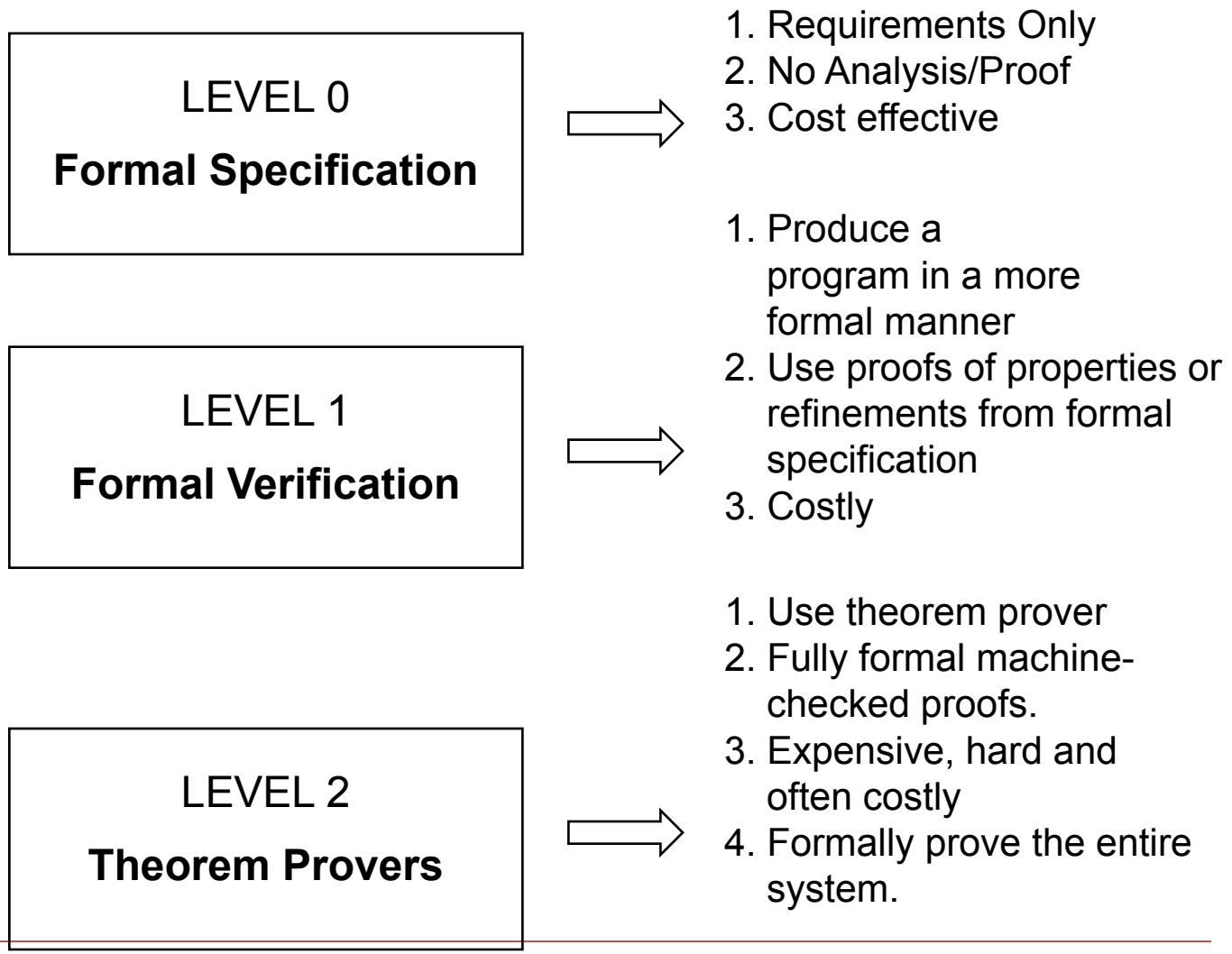
- Mathematical base allows precise notions.
- Unambiguous.
- Allows consistency, correctness, specification and implementation to be expressed.

Increase Human Understanding of Specified System.

&

Allow the possibility of formal reasoning and development.

Three Levels of Formal Methods



Thanks